

AI Tools Declaration

AI Tools Used:

ChatGPT (OpenAI)

Core Purpose:

The AI tool was used to assist in **refactoring the provided Python code** to improve its maintainability and to help structure the **analysis and report writing** for the assignment. It also provided guidance on reducing code duplication, improving modularity, and aligning the implementation with clean coding practices.

Prompts Shared:

- *“Refactor this Yahtzee Python code to improve maintainability while keeping the same functionality.”*
- *“Generate a maintainability report including comparison, problem identification, and refactoring explanation.”*

Critique & Validation:

The refactored code was tested to ensure it produced the **same functional output** as the original implementation. Additionally, static analysis tools (**radon**) were used to confirm improvements in **Cyclomatic Complexity and Maintainability Index**, ensuring the refactoring met the assignment requirements without introducing errors.